Triasha Sarkar

Machine Learning Engineer

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Professional Experience

Graduate Research Assistant, *Aerospace Systems Design Laboratory, Georgia Tech* May 2025 - Aug 2026 Atlanta, GA

- **M.S. Thesis - Scientific ML and Applied ML for Failure Detection in Time-Series Thrust Data:** Built and validated Python/scikit-learn workflows for simulation reliability and uncertainty-aware decision support, including data audit, preprocessing, feature engineering, and reproducible evaluation.
- Audited a **15,120-case** simulation campaign and identified 6,804 candidate simulator failures; removed duplicate-geometry leakage and trained a classifier with 96.8% accuracy and 0.986 ROC AUC on unseen geometries, then replicated results on an independent 1,050-case study.
- **HERO - Delta Air Lines-sponsored Applied ML project:** Helped build a RAG QA workflow over non-public FOQA/ASAP records to accelerate user-led safety and risk assessments. Contributed to source-aware organization that distinguished FOQA operational records from ASAP safety reports, supporting relevant, traceable retrieval for analysts' questions.
- **GREEN TEA - U.S. Endowment-sponsored Scientific and Applied ML project:** Co-developed a group model assessing forest-residue conversion to sustainable aviation fuel.Built a surrogate of Argonne National Laboratory's GREET life-cycle model; integrated it with economic and transportation models in a Tableau dashboard delivered to the U.S. Endowment for Forestry & Communities sponsor team.

Machine Learning Engineer, *Alten India, embedded at Rolls-Royce* Oct 2023 - Apr 2025 Bangalore, India

- Developed Python-based diagnostic and predictive-maintenance workflows for multivariate aircraft-engine sensor data, helping service engineers identify failure symptoms and prioritize component-health investigations.
- Ingested JSON, Parquet, HDF5, and CSV datasets from Azure Blob Storage using Azure Databricks; automated extraction and quality-control scripts and engineered 1 Hz/10 Hz time-series features for anomaly detection and component-health risk forecasting.
- Partnered with lifecycle engineers to define failure modes, validate diagnostic signals against engineering knowledge, and translate model outputs into actionable maintenance insights.
- Delivered pre-emptive engine-health reports that supported proactive maintenance planning and contributed to an estimated 5% reduction in maintenance cost.

Data Science Intern, *Rolls-Royce DataLab* May 2021 - Jul 2021 Bangalore, India

- Conducted EDA and feature engineering for predictive analytics, anomaly detection, performance-degradation analysis, and component-health monitoring in real-world aerospace datasets.
- Built proof-of-concept ML and model-based engineering workflows using time-series, sensor-fusion, and physics-informed approaches; collaborated with data engineers, domain experts, and senior data scientists to translate results into actionable insights.

Research Projects

AeroRAG-X, *Applied ML & AI/ML Engineering* GitHub Aug 2026

- **Designed and built AeroRAG-X, an evidence-grounded RAG system for NASA technical literature**, combining BM25 and dense retrieval, Reciprocal Rank Fusion, cross-encoder reranking, pgvector, evidence-sufficiency checks, citation-preserving generation, and structured outputs through FastAPI-based services.
- **Adapted a local Qwen LLM with PEFT/LoRA and built a protected evaluation framework for model and RAG quality**, improving conservative semantic concept coverage from **38.2% to 51.3%** and full answer-to-claim capture from **10% to 45%**, while maintaining **90%** broad claim-to-evidence support and **100%** safe non-assertion on grounded unsupported-query tests.
- **Implemented bounded adaptive retrieval with both native and LangGraph/LangChain orchestration**, using evidence assessment, conditional query rewriting, a maximum two-pass retrieval policy, deterministic termination, provenance-preserving refusal paths, and parity tests; evaluated adaptive versus single-pass RAG for recovery quality, unsupported assertions, citation integrity, latency, and retrieval overhead.
- **Hardened the platform for reproducible ML-system experimentation and deployment**, adding Dockerized services, CI/CD, OpenTelemetry/Prometheus observability, local inference and quantization benchmarks, multimodal text/table/figure retrieval, regression-tested evaluation suites, and versioned artifacts/configurations for reproducible releases

EdgeGenBench, *Scientific ML & AI/ML Engineering* GitHub Aug 2026

- Built a reproducible **6,000-sample synthetic** hybrid-electric and hydrogen aircraft-design benchmark; compared Ridge, Random Forest, HistGradientBoosting, and a multi-output PyTorch surrogate with training-only normalization and validation-based early stopping; v0.1 baselines reached 0.995 mean test R^2 and 99.89% ROC AUC.
- Measured held-out R^2, NRMSE, MAE, and RMSE with conformal uncertainty; added an **uncertainty gate** and false-safe-rate-aware feasibility classifier that defer risky designs to physics validation before constrained multi-objective optimization.

- Exported the neural surrogate to **ONNX FP32** and dynamically quantized INT8; validated prediction parity and benchmarked model size, P50/P95 latency, and throughput.
- Built and tested a **bounded Design Review Agent** with prediction, uncertainty, feasibility, optimization, physics-validation, and model-trade-off tools; enforced evidence-backed recommendations and a physics fallback, then packaged typed FastAPI/Docker APIs with integration tests.

NewsLens, *Applied ML & AI/ML Engineering* GitHub Jul 2026
- Built a chronological, leakage-aware evaluation pipeline over **5.84M candidate interactions** using Python, SQL, and DuckDB.
- Implemented TF-IDF personalization and a training-only popularity fallback; measured NDCG@10 0.3664, Recall@10 0.5955, and Hit Rate@10 0.6762.
- Added bootstrap confidence intervals plus subgroup and failure analysis to assess ranking behavior and failure modes.
- Served versioned artifacts through FastAPI with readiness, tracing, and latency reporting; maintained non-root Docker/Compose, multi-architecture GitHub Actions images, and **400+ automated tests**.

Technical Skills

Data and analytics: Python, SQL/MySQL, C++, MATLAB; pandas, NumPy, SciPy, DuckDB; Azure Blob Storage, Azure Databricks; EDA, feature engineering, statistical data analysis
Machine learning, time series, and GenAI: scikit-learn, PyTorch/TensorFlow, Hugging Face Transformers; predictive analytics, time-series analysis, anomaly detection; RAG, pgvector, BM25, dense retrieval, Sentence Transformers, RRF, cross-encoder reranking, grounded generation
Scientific ML and evaluation: surrogate modeling, uncertainty quantification, conformal prediction, constrained multi-objective optimization, physics validation; held-out evaluation, NDCG/MRR/Recall, ROC AUC, bootstrap confidence intervals
ML systems and cloud: FastAPI, REST APIs, Docker/Compose, Google Cloud Run, Cloud Storage, Azure DevOps, ONNX Runtime, Prometheus, OpenTelemetry, GitHub Actions, pytest, Ruff, mypy, CI/CD

Leadership & Service

Student Chair, *Women of Aerospace & Aeronautics (WOAA), Georgia Tech Chapter*
- Managed WOAA's social-media accounts and organized coffee chats for graduate students, coordinating chapter communications and creating an accessible forum for peer connection.

Education

Georgia Institute of Technology, *M.S., Aerospace Engineering* Aug 2026

Cranfield University, *M.Sc., Aerospace Vehicle Design* Feb 2023

SRM Institute of Science and Technology, *B.Tech., Aerospace Engineering* Jun 2021